1) a)

$\int_1^\infty \sqrt{x}\,\mathrm{d}x$ is an improper integral as one of the limits is ∞.

1) b)

$\int_0^1 \frac{1}{3x^3}\,\mathrm{d}x$ is an improper integral as $\frac{1}{3x^3}$ is undefined when $x = 0$, which is one of the limits.

2) a)

$$\int_2^\infty \frac{1}{x^2}\,\mathrm{d}x$$
$$= \lim_{a\to\infty} \int_2^a x^{-2}\,\mathrm{d}x$$
$$= \lim_{a\to\infty} \left[-\frac{1}{x}\right]_2^a$$
$$= \lim_{a\to\infty} \left(\left(-\frac{1}{a}\right) - \left(-\frac{1}{2}\right)\right)$$

As $a \to \infty$, $\frac{1}{a} \to 0$

$$\therefore \int_2^\infty \frac{1}{x^2}\,\mathrm{d}x = 0 - \left(-\frac{1}{2}\right) = \frac{1}{2}$$

2) b)

$$\int_2^\infty \frac{1}{x^3}\,\mathrm{d}x$$
$$= \lim_{a\to\infty} \int_2^a x^{-3}\,\mathrm{d}x$$
$$= \lim_{a\to\infty} \left[-\frac{1}{2x^2}\right]_2^a$$
$$= \lim_{a\to\infty} \left(\left(-\frac{1}{2a^2}\right) - \left(-\frac{1}{8}\right)\right)$$

As $a \to \infty$, $\frac{1}{a^2} \to 0$

$$\therefore \int_2^\infty \frac{1}{x^3}\,\mathrm{d}x = 0 - \left(-\frac{1}{8}\right) = \frac{1}{8}$$

2) c)

$\int_2^\infty \frac{1}{x^4}\,\mathrm{d}x$

$= \lim_{a\to\infty} \int_2^a x^{-4}\,\mathrm{d}x$

$= \lim_{a\to\infty} \left[-\frac{1}{3x^3}\right]_2^a$

$= \lim_{a\to\infty} \left(\left(-\frac{1}{3a^3}\right) - \left(-\frac{1}{24}\right)\right)$

As $a \to \infty$, $\frac{1}{a^2} \to 0$

$\therefore \int_2^\infty \frac{1}{x^4}\,\mathrm{d}x = 0 - \left(-\frac{1}{24}\right) = \frac{1}{24}$

3) a)

$\int_0^2 \frac{1}{x^2}\,\mathrm{d}x$

$= \lim_{a\to 0} \int_a^2 x^{-2}\,\mathrm{d}x$

$= \lim_{a\to 0} \left[-\frac{1}{x}\right]_a^2$

$= \lim_{a\to 0} \left(\left(-\frac{1}{2}\right) - \left(-\frac{1}{a}\right)\right)$

As $a \to 0$, $\frac{1}{a} \to \infty$

$\therefore \int_0^2 \frac{1}{x^2}\,\mathrm{d}x$ diverges

3) b)

$\int_a^2 \frac{1}{x^3}\,\mathrm{d}x$

$= \lim_{a\to 0} \int_a^2 x^{-3}\,\mathrm{d}x$

$= \lim_{a\to 0} \left[-\frac{1}{2x^2}\right]_a^2$

$= \lim_{a\to 0} \left(\left(-\frac{1}{8}\right) - \left(-\frac{1}{2a^2}\right)\right)$

As $a \to 0$, $\frac{1}{a^2} \to \infty$

$\therefore \int_0^2 \frac{1}{x^3}\,\mathrm{d}x$ diverges

3) c)

$\int_0^2 \frac{1}{x^4}\,\mathrm{d}x$

$= \lim_{a\to 0} \int_a^2 x^{-4}\,\mathrm{d}x$

$= \lim_{a\to 0} \left[-\frac{1}{3x^3}\right]_a^2$

$= \lim_{a\to 0} \left(\left(-\frac{1}{24}\right) - \left(-\frac{1}{3a^3}\right)\right)$

As $a \to 0$, $\frac{1}{a^3} \to \infty$

$\therefore \int_2^\infty \frac{1}{x^3}\,\mathrm{d}x$ diverges

4) a)

$\int_0^9 \frac{1}{\sqrt{x}}\,\mathrm{d}x$

$= \lim_{a\to 0} \int_a^9 x^{-\frac{1}{2}}\,\mathrm{d}x$

$= \lim_{a\to 0} \left[2\sqrt{x}\right]_a^9$

$= \lim_{a\to 0} \left(\left(2\sqrt{9}\right) - \left(2\sqrt{a}\right)\right)$

$= \lim_{a\to 0} \left(6 - 2\sqrt{a}\right)$

As $a \to 0$, $\sqrt{a} \to 0$

$\therefore \int_0^9 \frac{1}{\sqrt{x}}\,\mathrm{d}x = 6$

4) b)

$\int_4^\infty \frac{1}{\sqrt{x}}\,\mathrm{d}x$

$= \lim_{a\to \infty} \int_4^a x^{-\frac{1}{2}}\,\mathrm{d}x$

$= \lim_{a\to \infty} \left[2\sqrt{x}\right]_4^a$

$= \lim_{a\to \infty} \left(\left(2\sqrt{a}\right) - \left(2\sqrt{4}\right)\right)$

$= \lim_{a\to \infty} \left(2\sqrt{a} - 4\right)$

As $a \to \infty$, $\sqrt{a} \to \infty$

$\therefore \int_4^\infty \frac{1}{\sqrt{x}}\,\mathrm{d}x$ diverges

5) a)

$\int_0^8 \frac{1}{x^{\frac{4}{3}}}\,\mathrm{d}x$

$= \lim_{a\to 0} \int_a^8 x^{-\frac{4}{3}}\,\mathrm{d}x$

$= \lim_{a\to 0} \left[-3x^{-\frac{1}{3}}\right]_a^8$

$= \lim_{a\to 0} \left(\left(-3\times 8^{-\frac{1}{3}}\right) - \left(-3a^{-\frac{1}{3}}\right)\right)$

$= \lim_{a\to 0} \left(-\frac{3}{2} + 3a^{-\frac{1}{3}}\right)$

As $a \to 0$, $a^{-\frac{1}{3}} \to \infty$

$\therefore \int_0^8 \frac{1}{x^{\frac{4}{3}}}\,\mathrm{d}x$ diverges

5) b)

$\int_8^\infty \frac{1}{x^{\frac{4}{3}}}\,\mathrm{d}x$

$= \lim_{a\to \infty} \int_8^a x^{-\frac{4}{3}}\,\mathrm{d}x$

$= \lim_{a\to \infty} \left[-3x^{-\frac{1}{3}}\right]_8^a$

$= \lim_{a\to \infty} \left(\left(-3a^{-\frac{1}{3}}\right) - \left(-3\times 8^{-\frac{1}{3}}\right)\right)$

$= \lim_{a\to \infty} \left(-3a^{-\frac{1}{3}} + \frac{3}{2}\right)$

As $a \to \infty$, $a^{-\frac{1}{3}} \to 0$

$\therefore \int_8^\infty \frac{1}{x^{\frac{4}{3}}}\,\mathrm{d}x = \frac{3}{2}$

6) a)

$\int_4^\infty 192x^{-4}\,\mathrm{d}x$

$= \lim_{a\to\infty} \int_4^a 192x^{-4}\,\mathrm{d}x$

$= \lim_{a\to\infty} [-64x^{-3}]_4^a$

$= \lim_{a\to\infty} \left((-64a^{-3}) - (-64\times 4^{-3})\right)$

$= \lim_{a\to\infty} (-64a^{-3} + 1)$

As $a \to \infty$, $a^{-3} \to 0$

$\therefore \int_4^\infty 192x^{-4}\,\mathrm{d}x = 1$

6) b)

$\int_4^\infty (192x^{-4} + 1)\,\mathrm{d}x$

$= \lim_{a\to\infty} \int_4^a (192x^{-4} + 1)\,\mathrm{d}x$

$= \lim_{a\to\infty} [-64x^{-3} + x]_4^a$

$= \lim_{a\to\infty} \left((-64a^{-3} + a) - (-64\times 4^{-3} + 4)\right)$

$= \lim_{a\to\infty} (-64a^{-3} + a - 3)$

As $a \to \infty$, $a^{-3} \to 0$

$\therefore \int_4^\infty (192x^{-4} + 1)\,\mathrm{d}x$ diverges

6) c)

$\int_4^\infty 192x^{-4}(x+1)\,\mathrm{d}x$

$= \lim_{a\to\infty} \int_4^a (192x^{-3} + 192x^{-4})\,\mathrm{d}x$

$= \lim_{a\to\infty} [-96x^{-2} - 64x^{-3}]_4^a$

$= \lim_{a\to\infty} \left((-96a^{-2} - 64a^{-3}) - (-96\times 4^{-2} - 64\times 4^{-3})\right)$

$= \lim_{a\to\infty} (-96a^{-2} - 64a^{-3} + 7)$

As $a \to \infty$, $a^{-2} \to 0$ and $a^{-3} \to 0$

$\therefore \int_4^\infty 192x^{-4}(x+1)\,\mathrm{d}x = 7$

7)

$\int_0^4 \frac{1}{x^3\sqrt{x}}\,\mathrm{d}x$

$= \lim_{a\to 0} \int_a^4 \frac{1}{x^{\frac{7}{2}}}\,\mathrm{d}x$

$= \lim_{a\to 0} \int_a^4 x^{-\frac{7}{2}}\,\mathrm{d}x$

$= \lim_{a\to 0} \left[-\frac{2}{5}x^{-\frac{5}{2}}\right]_a^4$

$= \lim_{a\to 0} \left(\left(-\frac{2}{5}\times 4^{-\frac{5}{2}}\right) - \left(-\frac{2}{5}a^{-\frac{5}{2}}\right)\right)$

$= \lim_{a\to 0} \left(-\frac{1}{80} + \frac{2}{5}a^{-\frac{5}{2}}\right)$

As $a \to 0$, $a^{-\frac{5}{2}} \to \infty$

$\therefore \int_0^4 \frac{1}{x^3\sqrt{x}}\,\mathrm{d}x$ diverges

8)

$\int_0^{\frac{1}{64}} \left(\frac{1}{\sqrt{x}} - \frac{1}{\sqrt[3]{x}}\right)\mathrm{d}x$

$= \lim_{a\to 0} \int_a^{\frac{1}{64}} \left(x^{-\frac{1}{2}} - x^{-\frac{1}{3}}\right)\mathrm{d}x$

$= \lim_{a\to 0} \left[2x^{\frac{1}{2}} - \frac{3}{2}x^{\frac{2}{3}}\right]_a^{\frac{1}{64}}$

$= \lim_{a\to 0} \left(\left(2\times\left(\frac{1}{64}\right)^{\frac{1}{2}} - \frac{3}{2}\times\left(\frac{1}{64}\right)^{\frac{2}{3}}\right) - \left(2a^{\frac{1}{2}} - \frac{3}{2}a^{\frac{2}{3}}\right)\right)$

$= \lim_{a\to 0} \left(\frac{5}{32} - 2a^{\frac{1}{2}} + \frac{3}{2}a^{\frac{2}{3}}\right)$

As $a \to 0$, $a^{\frac{1}{2}} \to 0$ and $a^{\frac{2}{3}} \to 0$

$\therefore \int_0^{\frac{1}{64}} \left(\frac{1}{\sqrt{x}} - \frac{1}{\sqrt[3]{x}}\right)\mathrm{d}x = \frac{5}{32}$

9) a)

Amelia's method is incorrect as $\frac{35}{x^6}$ is undefined when $x = 0$, and hence the integral is improper. Amelia should instead split the integral up into two integrals, where one has the limits from -2 to a, and the other with the limits a to 2, and then evaluate the integrals by considering the limit as a tends to zero.

9) b)

$\int_{-2}^{2} \frac{35}{x^6}\,\mathrm{d}x = \int_{-2}^{0} \frac{35}{x^6}\,\mathrm{d}x + \int_{0}^{2} \frac{35}{x^6}\,\mathrm{d}x$

$= \lim_{a\to 0} \int_{-2}^{a} 35x^{-6}\,\mathrm{d}x + \lim_{a\to 0} \int_{a}^{2} 35x^{-6}\,\mathrm{d}x$

$= \lim_{a\to 0} [-7x^{-5}]_{-2}^{a} + \lim_{a\to 0} [-7x^{-5}]_{a}^{2}$

$= \lim_{a\to 0} \left((-7a^{-5}) - (-7(-2)^{-5})\right) + \lim_{a\to 0} \left((-7(2)^{-5}) - (-7a^{-5})\right)$

$= \lim_{a\to 0} \left(-7a^{-5} - \frac{7}{32}\right) + \lim_{a\to 0} \left(-\frac{7}{32} + 7a^{-5}\right)$

As $a \to 0$, $a^{-5} \to \infty$

$\therefore \int_{-2}^{2} \frac{35}{x^6}\,\mathrm{d}x$ diverges

10)

$y = \frac{4}{x^{\frac{2}{3}}}$ is undefined when $x = 0$

$\therefore \int_{-1}^{8} \frac{4}{x^{\frac{2}{3}}}\,\mathrm{d}x = \int_{-1}^{0} \frac{4}{x^{\frac{2}{3}}}\,\mathrm{d}x + \int_{0}^{8} \frac{4}{x^{\frac{2}{3}}}\,\mathrm{d}x$

$= \lim_{a\to 0} \int_{-1}^{a} 4x^{-\frac{2}{3}}\,\mathrm{d}x + \lim_{a\to 0} \int_{a}^{8} 4x^{-\frac{2}{3}}\,\mathrm{d}x$

$= \lim_{a\to 0} \left[12x^{\frac{1}{3}}\right]_{-1}^{a} + \lim_{a\to 0} \left[12x^{\frac{1}{3}}\right]_{a}^{8}$

$= \lim_{a\to 0} \left(\left(12a^{\frac{1}{3}}\right) - \left(12(-1)^{\frac{1}{3}}\right)\right) + \lim_{a\to 0} \left(\left(12 \times 8^{\frac{1}{3}}\right) - \left(12a^{\frac{1}{3}}\right)\right)$

$= \lim_{a\to 0} \left(12a^{\frac{1}{3}} + 12\right) + \lim_{a\to 0} \left(24 - 12a^{\frac{1}{3}}\right)$

As $a \to 0$, $a^{\frac{1}{3}} \to 0$

$\therefore \int_{-1}^{8} \frac{4}{x^{\frac{2}{3}}}\,\mathrm{d}x = 12 + 24 = 36$

11)

$y = \frac{100}{x^{\frac{4}{5}}}$ is undefined when $x = 0$

$\therefore \int_{-32}^{1} \frac{100}{x^{\frac{4}{5}}}\,\mathrm{d}x = \int_{-32}^{0} \frac{100}{x^{\frac{4}{5}}}\,\mathrm{d}x + \int_{0}^{1} \frac{100}{x^{\frac{4}{5}}}\,\mathrm{d}x$

$= \lim_{a\to 0} \int_{-32}^{a} 100x^{-\frac{4}{5}}\,\mathrm{d}x + \lim_{a\to 0} \int_{a}^{1} 100x^{-\frac{4}{5}}\,\mathrm{d}x$

$= \lim_{a\to 0} \left[500x^{\frac{1}{5}}\right]_{-32}^{a} + \lim_{a\to 0} \left[500x^{\frac{1}{5}}\right]_{a}^{1}$

$= \lim_{a\to 0}\left(\left(500a^{\frac{1}{5}}\right) - \left(500\times(-32)^{\frac{1}{5}}\right)\right) + \lim_{a\to 0}\left(\left(500\times 1^{\frac{1}{5}}\right) - \left(500a^{\frac{1}{5}}\right)\right)$

$= \lim_{a\to 0}\left(500a^{\frac{1}{5}} + 1000\right) + \lim_{a\to 0}\left(500 - 500a^{\frac{1}{5}}\right)$

As $a \to 0$, $a^{\frac{1}{5}} \to 0$

$\therefore \int_{-32}^{1} \frac{100}{x^{\frac{4}{5}}}\,\mathrm{d}x = 1000 + 500 = 1500$

12) a)

$\int_4^6 \frac{1}{x-5}\,\mathrm{d}x$ is an improper integral as $\frac{1}{x-5}$ is undefined when $x = 5$, which is between the limits $x = 4$ and $x = 6$.

12) b)

$\int_0^1 \frac{\mathrm{e}^x}{\mathrm{e}^x-2}\,\mathrm{d}x$ is an improper integral as $\frac{\mathrm{e}^x}{\mathrm{e}^x-2}$ is undefined when $\mathrm{e}^x - 2 = 0 \Rightarrow \mathrm{e}^x = 2$ $\Rightarrow x = \ln 2 \approx 0.693$, which is between the limits $x = 0$ and $x = 1$.

12) c)

$\int_0^{\frac{2\pi}{3}} \frac{\sqrt{3}\sin x+\cos x}{\sin x-\sqrt{3}\cos x}\,\mathrm{d}x$ is an improper integral as $\frac{\sqrt{3}\sin x+\cos x}{\sin x-\sqrt{3}\cos x}$ is undefined when $\sin x - \sqrt{3}\cos x = 0 \Rightarrow \sin x = \sqrt{3}\cos x \Rightarrow \frac{\sin x}{\cos x} = \sqrt{3} \Rightarrow \tan x = \sqrt{3} \Rightarrow x = \frac{\pi}{3}$, which is between the limits $x = 0$ and $x = \frac{2\pi}{3}$.

12) d)

$\int_0^1 x^4 \ln x\,\mathrm{d}x$ is an improper integral as $\ln x$ is undefined when $x = 0$, which is one of the limits.

13) a)

$\int_2^\infty \frac{1}{x}\,\mathrm{d}x$

$= \lim_{a\to\infty} \int_2^a \frac{1}{x}\,\mathrm{d}x$

$= \lim_{a\to\infty} [\ln|x|]_2^a$

$= \lim_{a\to\infty} (\ln|a| - \ln 2)$

As $a \to \infty$, $\ln|a| \to \infty$

$\therefore \int_2^\infty \frac{1}{x}\,\mathrm{d}x$ diverges

13) b)

$\int_0^2 \frac{1}{x}\,\mathrm{d}x$

$= \lim_{a\to 0} \int_a^2 \frac{1}{x}\,\mathrm{d}x$

$= \lim_{a\to 0} [\ln|x|]_a^2$

$= \lim_{a\to 0} (\ln 2 - \ln|a|)$

As $a \to 0$, $\ln|a| \to -\infty$

$\therefore \int_0^2 \frac{1}{x}\,\mathrm{d}x$ diverges

14) a)

$\int_0^\infty \frac{1}{\mathrm{e}^x}\,\mathrm{d}x$

$= \lim_{a\to\infty} \int_0^a \mathrm{e}^{-x}\,\mathrm{d}x$

$= \lim_{a\to\infty} [-\mathrm{e}^{-x}]_0^a$

$= \lim_{a\to\infty} \left((-\mathrm{e}^{-a}) - (-\mathrm{e}^0)\right)$

$= \lim_{a\to\infty} (-\mathrm{e}^{-a} + 1)$

As $a \to \infty$, $\mathrm{e}^{-a} \to 0$

$\therefore \int_0^\infty \frac{1}{\mathrm{e}^x}\,\mathrm{d}x = 1$

14) b)

$\int_0^\infty \frac{1}{e^{2x}}\,dx$

$= \lim_{a\to\infty} \int_0^a e^{-2x}\,dx$

$= \lim_{a\to\infty} \left[-\frac{1}{2}e^{-2x}\right]_0^a$

$= \lim_{a\to\infty} \left(\left(-\frac{1}{2}e^{-2a}\right) - \left(-\frac{1}{2}e^{0}\right)\right)$

$= \lim_{a\to\infty} \left(-\frac{1}{2}e^{-2a} + \frac{1}{2}\right)$

As $a \to \infty$, $e^{-2a} \to 0$

$\therefore \int_0^\infty \frac{1}{e^{2x}}\,dx = \frac{1}{2}$

15)

$\int_7^\infty (x-3)^{-\frac{5}{2}}\,dx$

$= \lim_{a\to\infty} \int_7^a (x-3)^{-\frac{5}{2}}\,dx$

$= \lim_{a\to\infty} \left[-\frac{2}{3}(x-3)^{-\frac{3}{2}}\right]_7^a$

$= \lim_{a\to\infty} \left(\left(-\frac{2}{3}(a-3)^{-\frac{3}{2}}\right) - \left(-\frac{2}{3}(7-3)^{-\frac{3}{2}}\right)\right)$

$= \lim_{a\to\infty} \left(-\frac{2}{3}(a-3)^{-\frac{3}{2}} + \frac{1}{12}\right)$

As $a \to \infty$, $(a-3)^{-\frac{3}{2}} \to 0$

$\therefore \int_7^\infty (x-3)^{-\frac{5}{2}}\,dx = \frac{1}{12}$

16)

$y = \frac{2}{(2x+3)^{\frac{6}{7}}}$ is undefined when $x = 0$

$\therefore \int_{-2}^{62.5} \frac{2}{(2x+3)^{\frac{6}{7}}}\,\mathrm{d}x = \int_{-2}^{-1.5} \frac{2}{(2x+3)^{\frac{6}{7}}}\,\mathrm{d}x + \int_{-1.5}^{62.5} \frac{2}{(2x+3)^{\frac{6}{7}}}\,\mathrm{d}x$

$= \lim_{a\to -1.5} \int_{-2}^{a} 2(2x+3)^{-\frac{6}{7}}\,\mathrm{d}x + \lim_{a\to -1.5} \int_{a}^{62.5} 2(2x+3)^{-\frac{6}{7}}\,\mathrm{d}x$

$= \lim_{a\to -1.5} \left[7(2x+3)^{\frac{1}{7}}\right]_{-2}^{a} + \lim_{a\to -1.5} \left[7(2x+3)^{\frac{1}{7}}\right]_{a}^{62.5}$

$= \lim_{a\to -1.5} \left(\left(7(2a+3)^{\frac{1}{7}}\right) - \left(7(2\times(-2)+3)^{\frac{1}{7}}\right)\right)$

$+ \lim_{a\to -1.5} \left(\left(7(2\times 62.5+3)^{\frac{1}{7}}\right) - \left(7(2a+3)^{\frac{1}{7}}\right)\right)$

$= \lim_{a\to -1.5} \left(7(2a+3)^{\frac{1}{7}} + 7\right) + \lim_{a\to -1.5} \left(14 - 7(2a+3)^{\frac{1}{7}}\right)$

As $a \to -1.5$, $7(2a+3)^{\frac{1}{7}} \to 0$

$\therefore \int_{-2}^{62.5} \frac{2}{(2x+3)^{\frac{6}{7}}}\,\mathrm{d}x = 7 + 14 = 21$

17) a)

$\int_{3}^{4} \frac{x}{x^2-9}\,\mathrm{d}x$

$= \lim_{a\to 3} \int_{a}^{4} \frac{x}{x^2-9}\,\mathrm{d}x$

$= \lim_{a\to 3} \left[\frac{1}{2}\ln|x^2-9|\right]_{a}^{4}$

$= \lim_{a\to 3} \left(\frac{1}{2}\ln|4^2-9| - \frac{1}{2}\ln|a^2-9|\right)$

As $a \to 3$, $\ln|a^2-9| \to -\infty$

$\therefore \int_{3}^{4} \frac{x}{x^2-9}\,\mathrm{d}x$ diverges

17) b)

$\int_3^4 \frac{x}{\sqrt{x^2-9}}\,\mathrm{d}x$

$= \lim_{a\to 3} \int_a^4 \frac{x}{\sqrt{x^2-9}}\,\mathrm{d}x$

$= \lim_{a\to 3} \left[\sqrt{x^2-9}\right]_a^4$

$= \lim_{a\to 3} \left(\sqrt{4^2-9} - \sqrt{a^2-9}\right)$

$= \lim_{a\to 3} \left(\sqrt{7} - \sqrt{a^2-9}\right)$

As $a \to 3$, $\sqrt{a^2-9} \to 0$

$\therefore \int_3^4 \frac{x}{\sqrt{x^2-9}}\,\mathrm{d}x = \sqrt{7}$

18) a)

$\int_0^\infty 4x\mathrm{e}^{-x^2}\,\mathrm{d}x$

$= \lim_{a\to\infty} \int_0^a 4x\mathrm{e}^{-x^2}\,\mathrm{d}x$

$= \lim_{a\to\infty} \left[-2\mathrm{e}^{-x^2}\right]_0^a$

$= \lim_{a\to\infty} \left(\left(-2\mathrm{e}^{-a^2}\right) - \left(-2\mathrm{e}^{-0^2}\right)\right)$

$= \lim_{a\to\infty} \left(-2\mathrm{e}^{-a^2} + 2\right)$

As $a \to \infty$, $\mathrm{e}^{-a^2} \to 0$

$\therefore \int_0^\infty 4x\mathrm{e}^{-x^2}\,\mathrm{d}x = 2$

18) b)

$\int_{-\infty}^\infty 4x\mathrm{e}^{-x^2}\,\mathrm{d}x$

$= \int_{-\infty}^0 4x\mathrm{e}^{-x^2}\,\mathrm{d}x + \int_0^\infty 4x\mathrm{e}^{-x^2}\,\mathrm{d}x$

$= \lim_{a\to-\infty} \int_a^0 4x\mathrm{e}^{-x^2}\,\mathrm{d}x + \lim_{a\to\infty} \int_0^a 4x\mathrm{e}^{-x^2}\,\mathrm{d}x$

$= \lim_{a\to-\infty} \left[-2\mathrm{e}^{-x^2}\right]_a^0 + \lim_{a\to\infty} \left[-2\mathrm{e}^{-x^2}\right]_0^a$

$= \lim_{a\to-\infty} \left(\left(-2\mathrm{e}^{-0^2}\right) - \left(-2\mathrm{e}^{-a^2}\right)\right) + \lim_{a\to\infty} \left(\left(-2\mathrm{e}^{-a^2}\right) - \left(-2\mathrm{e}^{-0^2}\right)\right)$

$= \lim_{a\to-\infty} \left(-2 + 2\mathrm{e}^{-a^2}\right) + \lim_{a\to\infty} \left(-2\mathrm{e}^{-a^2} + 2\right)$

As $a \to \infty$, $\mathrm{e}^{-a^2} \to 0$ and as $a \to -\infty$, $\mathrm{e}^{-a^2} \to 0$

$\therefore \int_{-\infty}^\infty 4x\mathrm{e}^{-x^2}\,\mathrm{d}x = -2 + 2 = 0$

19)

$\int_{-\infty}^{0} \frac{e^x}{e^x+1}\,dx$

$= \lim_{a\to-\infty} \int_{a}^{0} \frac{e^x}{e^x+1}\,dx$

$= \lim_{a\to-\infty} [\ln|e^x+1|]_a^0$

$= \lim_{a\to-\infty} (\ln|e^0+1| - \ln|e^a+1|)$

$= \lim_{a\to-\infty} (\ln 2 - \ln|e^a+1|)$

As $a \to -\infty$, $e^a \to 0$

$\therefore \int_{-\infty}^{0} \frac{e^x}{e^x+1}\,dx = \ln 2$

20)

$\int_{0}^{\frac{\pi}{2}} \tan x\,dx$

$= \lim_{a\to\frac{\pi}{2}} \int_{0}^{a} \tan x\,dx$

$= \lim_{a\to\frac{\pi}{2}} [-\ln|\cos x|]_0^a$

$= \lim_{a\to\frac{\pi}{2}} \big((-\ln|\cos a|) - (-\ln|\cos 0|)\big)$

$= \lim_{a\to\frac{\pi}{2}} (-\ln|\cos a| + 0)$

As $a \to \frac{\pi}{2}$, $\ln|\cos a| \to -\infty$

$\therefore \int_{0}^{\frac{\pi}{2}} \tan x\,dx$ diverges

21)

$\int_{-1}^{2} \frac{x+1}{\sqrt{8-2x-x^2}}\,dx$

$= \lim_{a\to2} \int_{-1}^{a} \frac{x+1}{\sqrt{8-2x-x^2}}\,dx$

$= \lim_{a\to2} \left[-\sqrt{8-2x-x^2}\right]_{-1}^{a}$

$= \lim_{a\to2} \left(\left(-\sqrt{8-2a-a^2}\right) - \left(-\sqrt{8-2(-1)-(-1)^2}\right)\right)$

$= \lim_{a\to2} \left(-\sqrt{8-2a-a^2}+3\right)$

As $a \to 2$, $\sqrt{8-2a-a^2} \to 0$

$\therefore \int_{-1}^{2} \frac{x+1}{\sqrt{8-2x-x^2}}\,dx = 3$

22) a)

$\lim_{a\to\infty}\left(\frac{6a-5}{3a+7}\right)=\lim_{a\to\infty}\left(\frac{6-\frac{5}{a}}{3+\frac{7}{a}}\right)=\frac{6}{3}=2$

As $a\to\infty$, $\frac{5}{a}\to 0$ and $\frac{7}{a}\to 0$

22) b)

$\int_4^\infty\left(\frac{6}{6x-5}-\frac{3}{3x+7}\right)\mathrm{d}x$

$=\lim_{a\to\infty}\int_4^a\left(\frac{6}{6x-5}-\frac{3}{3x+7}\right)\mathrm{d}x$

$=\lim_{a\to\infty}[\ln|6x-5|-\ln|3x+7|]_4^a$

$=\lim_{a\to\infty}\left((\ln|6a-5|-\ln|3a+7|)-(\ln|19|-\ln|19|)\right)$

$=\lim_{a\to\infty}\left(\ln\left|\frac{6a-5}{3a+7}\right|\right)$

As $a\to\infty$, $\frac{6a-5}{3a+7}\to 2$

$\therefore\int_4^\infty\left(\frac{6}{6x-5}-\frac{3}{3x+7}\right)\mathrm{d}x=\ln 2$

23)

$\frac{2}{x(3x+1)} \equiv \frac{A}{x} + \frac{B}{3x+1}$

$\Rightarrow 2 \equiv A(3x+1) + Bx$

Let $x = 0$: $2 = A$

Let $x = -\frac{1}{3}$: $2 = -\frac{1}{3}B \Rightarrow B = -6$

$\therefore \frac{2}{x(3x+1)} \equiv \frac{2}{x} - \frac{6}{3x+1}$

$\int_1^\infty \frac{3x-2}{x(3x+1)}\,\mathrm{d}x = \int_1^\infty \left(\frac{2}{x} - \frac{6}{3x+1}\right)\mathrm{d}x$

$= \lim_{a\to\infty} \int_1^a \left(\frac{2}{x} - \frac{6}{3x+1}\right)\mathrm{d}x$

$= \lim_{a\to\infty} [2\ln|x| - 2\ln|3x+1|]_1^a$

$= \lim_{a\to\infty} \big((2\ln|a| - 2\ln|3a+1|) - (2\ln|1| - 2\ln|3\times 1+1|)\big)$

$= \lim_{a\to\infty} (2(\ln|a| - \ln|3a+1|) + 2\ln 4)$

$= \lim_{a\to\infty} \left(2\ln\left|\frac{a}{3a+1}\right| + 2\ln 4\right)$

As $a \to \infty$, $\frac{a}{3a+1} = \frac{1}{3+\frac{1}{a}} \to \frac{1}{3}$

$\therefore \int_1^\infty \frac{3x-2}{x(3x+1)}\,\mathrm{d}x = 2\ln\frac{1}{3} + 2\ln 4$

$= 2\left(\ln\frac{1}{3} + \ln 4\right)$

$= 2\ln\frac{4}{3}$

$= \ln\frac{16}{9}$

i.e. $k = \frac{16}{9}$

24)

$\int_{-1}^{2} \frac{2-x}{\sqrt{x+1}} \mathrm{d}x$

Use Integration by Substitution

Let $u = x + 1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 1 \Rightarrow \mathrm{d}u = \mathrm{d}x$

Change limits:

$x = 2 \Rightarrow u = 3$

$x = -1 \Rightarrow u = 0$

$u = x + 1 \Rightarrow x = u - 1 \Rightarrow 2 - x = 2 - (u - 1) = 3 - u$

$\therefore \int_{-1}^{2} \frac{2-x}{\sqrt{x+1}} \mathrm{d}x = \int_{0}^{3} \frac{3-u}{\sqrt{u}} \mathrm{d}u$

$= \lim_{a \to 0} \int_{a}^{3} \frac{3-u}{\sqrt{u}} \mathrm{d}u$

$= \lim_{a \to 0} \int_{a}^{3} \left(3u^{-\frac{1}{2}} - u^{\frac{1}{2}}\right) \mathrm{d}u$

$= \lim_{a \to 0} \left[6u^{\frac{1}{2}} - \frac{2}{3}u^{\frac{3}{2}}\right]_{a}^{3}$

$= \lim_{a \to 0} \left(\left(6 \times 3^{\frac{1}{2}} - \frac{2}{3} \times 3^{\frac{3}{2}}\right) - \left(6a^{\frac{1}{2}} - \frac{2}{3}a^{\frac{3}{2}}\right)\right)$

$= \lim_{a \to 0} \left(4\sqrt{3} - 6a^{\frac{1}{2}} + \frac{2}{3}a^{\frac{3}{2}}\right)$

As $a \to 0$, $a^{\frac{1}{2}} \to 0$ and $a^{\frac{3}{2}} \to 0$

$\therefore \int_{-1}^{2} \frac{2-x}{\sqrt{x+1}} \mathrm{d}x = 4\sqrt{3}$

25)

$\int_{\frac{1}{\pi}}^{\infty} \frac{\sin\left(\frac{1}{x}\right)}{x^2}\,\mathrm{d}x$

$= \lim_{a\to\infty} \int_{\frac{1}{\pi}}^{a} \frac{\sin\left(\frac{1}{x}\right)}{x^2}\,\mathrm{d}x$

$= \lim_{a\to\infty} \left[\cos\left(\frac{1}{x}\right)\right]_{\frac{1}{\pi}}^{a}$

$= \lim_{a\to\infty} \left(\cos\left(\frac{1}{a}\right) - \cos\left(\frac{1}{\frac{1}{\pi}}\right)\right)$

$= \lim_{a\to\infty} \left(\cos\left(\frac{1}{a}\right) - \cos\pi\right)$

$= \lim_{a\to\infty} \left(\cos\left(\frac{1}{a}\right) + 1\right)$

As $a \to \infty$, $\frac{1}{a} \to 0$ so $\cos\left(\frac{1}{a}\right) \to 1$

$\therefore \int_{\frac{1}{\pi}}^{\infty} \frac{\sin\left(\frac{1}{x}\right)}{x^2}\,\mathrm{d}x = 1 + 1 = 2$

26)

$\int_{\frac{3}{2}}^{\infty} \frac{(3-2x)x^2}{x^3-2e^{2x}}\,dx = \lim_{a\to\infty} \int_{\frac{3}{2}}^{a} \frac{(3-2x)x^2}{x^3-2e^{2x}}\,dx$

Use Integration by Substitution:

$u = 2 - x^3e^{-2x}$

$\Rightarrow \frac{du}{dx} = -x^3 \times -2e^{-2x} - 3x^2 \times e^{-2x}$

$\Rightarrow \frac{du}{dx} = 2x^3e^{-2x} - 3x^2e^{-2x}$

$\Rightarrow \frac{du}{dx} = (2x-3)x^2e^{-2x}$

$\Rightarrow dx = \frac{1}{(2x-3)x^2e^{-2x}}\,du$

$\Rightarrow dx = \frac{e^{2x}}{(2x-3)x^2}\,du$

$\int \frac{(3-2x)x^2}{x^3-2e^{2x}}\,dx = \int \frac{(3-2x)x^2}{x^3-2e^{2x}} \times \frac{e^{2x}}{(2x-3)x^2}\,du = \int -\frac{e^{2x}}{x^3-2e^{2x}}\,du$

$= \int -\frac{1}{x^3e^{-2x}-2}\,du = \int \frac{1}{2-x^3e^{-2x}}\,du = \int \frac{1}{u}\,du$

Change limits:

$x = a \Rightarrow u = 2 - a^3e^{-2a}$

$x = \frac{3}{2} \Rightarrow u = 2 - \left(\frac{3}{2}\right)^3 e^{-2\times\frac{3}{2}} = 2 - \frac{27}{8}e^{-3}$

$\therefore \lim_{a\to\infty} \int_{\frac{3}{2}}^{a} \frac{(3-2x)x^2}{x^3-2e^{2x}}\,dx = \lim_{a\to\infty} \int_{2-\frac{27}{8}e^{-3}}^{2-a^3e^{-2a}} \frac{1}{u}\,du$

$= \lim_{a\to\infty} [\ln|u|]_{2-\frac{27}{8}e^{-3}}^{2-a^3e^{-2a}}$

$= \lim_{a\to\infty} \left(\ln|2 - a^3e^{-2a}| - \ln\left|2 - \frac{27}{8}e^{-3}\right|\right)$

As $a \to \infty$, $a^3e^{-2a} \to 0$

$\therefore \int_{\frac{3}{2}}^{\infty} \frac{(3-2x)x^2}{x^3-2e^{2x}}\,dx = \ln 2 - \ln\left(2 - \frac{27}{8}e^{-3}\right)$

$= \ln\left(\frac{2}{2-\frac{27}{8}e^{-3}}\right)$

$= \ln\left(\frac{16e^3}{16e^3-27}\right)$

27)

$\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x = \int_{-1}^{0} \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x + \int_{0}^{1} \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x$

$= \lim_{a\to -1} \int_{a}^{0} \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x + \lim_{a\to 1} \int_{0}^{a} \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x$

$= \lim_{a\to -1} [\arcsin x]_{a}^{0} + \lim_{a\to 1} [\arcsin x]_{0}^{a}$

$= \lim_{a\to -1} (\arcsin 0 - \arcsin a) + \lim_{a\to 1} (\arcsin a - \arcsin 0)$

$= \lim_{a\to -1} (-\arcsin a) + \lim_{a\to 1} (\arcsin a)$

As $a \to -1$, $\arcsin a \to -\frac{\pi}{2}$

As $a \to 1$, $\arcsin a \to \frac{\pi}{2}$

$\therefore \int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x = -\left(-\frac{\pi}{2}\right) + \frac{\pi}{2}$

$= \pi$

28)

$\int_{0}^{\infty} (\cosh x - \sinh x)\,\mathrm{d}x$

$= \lim_{a\to \infty} \int_{0}^{a} (\cosh x - \sinh x)\,\mathrm{d}x$

$= \lim_{a\to \infty} [\sinh x - \cosh x]_{0}^{a}$

$= \lim_{a\to \infty} \left[\frac{\mathrm{e}^{x} - \mathrm{e}^{-x}}{2} - \frac{\mathrm{e}^{x} + \mathrm{e}^{-x}}{2}\right]_{0}^{a}$

$= \lim_{a\to \infty} [-\mathrm{e}^{-x}]_{0}^{a}$

$= \lim_{a\to \infty} \left((-\mathrm{e}^{-a}) - (-\mathrm{e}^{0})\right)$

$= \lim_{a\to \infty} (-\mathrm{e}^{-a} + 1)$

As $a \to \infty$, $\mathrm{e}^{-a} \to 0$

$\therefore \int_{0}^{\infty} (\cosh x - \sinh x)\,\mathrm{d}x = 1$

29) a)

$\int_2^\infty \frac{1}{4x^2-9}\,\mathrm{d}x$

$\frac{1}{4x^2-9} \equiv \frac{1}{(2x-3)(2x+3)} \equiv \frac{A}{2x-3} + \frac{B}{2x+3}$

$\Rightarrow 1 \equiv A(2x+3) + B(2x-3)$

Let $x = \frac{3}{2}$: $1 = 6A \Rightarrow A = \frac{1}{6}$

Let $x = -\frac{3}{2}$: $1 = -6B \Rightarrow B = -\frac{1}{6}$

$\therefore \frac{1}{4x^2-9} \equiv \frac{\frac{1}{6}}{2x-3} - \frac{\frac{1}{6}}{2x+3}$

$\therefore \int_2^\infty \frac{1}{4x^2-9}\,\mathrm{d}x$

$= \lim_{a\to\infty} \int_2^a \left(\frac{\frac{1}{6}}{2x-3} - \frac{\frac{1}{6}}{2x+3}\right)\mathrm{d}x$

$= \lim_{a\to\infty} \left[\frac{1}{12}\ln|2x-3| - \frac{1}{12}\ln|2x+3|\right]_2^a$

$= \lim_{a\to\infty} \left(\left(\frac{1}{12}\ln|2a-3| - \frac{1}{12}\ln|2a+3|\right) - \left(\frac{1}{12}\ln|1| - \frac{1}{12}\ln|7|\right)\right)$

$= \lim_{a\to\infty} \left(\frac{1}{12}(\ln|2a-3| - \ln|2a+3|) + \frac{1}{12}\ln 7\right)$

$= \lim_{a\to\infty} \left(\frac{1}{12}\ln\left|\frac{2a-3}{2a+3}\right| + \frac{1}{12}\ln 7\right)$

As $a \to \infty$, $\frac{2a-3}{2a+3} = \frac{2-\frac{3}{a}}{2+\frac{3}{a}} \to 1$, so $\ln\left(\frac{2a-3}{2a+3}\right) \to 0$

$\therefore \int_2^\infty \frac{1}{4x^2-9}\,\mathrm{d}x = \frac{1}{12}\ln 7$

29) b)

$\int_0^\infty \frac{1}{4x^2+9}\,\mathrm{d}x$

$= \lim_{a\to\infty} \int_0^a \frac{1}{9\left(\frac{4}{9}x^2+1\right)}\,\mathrm{d}x$

$= \lim_{a\to\infty} \int_0^a \frac{1}{9} \times \frac{1}{1+\left(\frac{2}{3}x\right)^2}\,\mathrm{d}x$

$= \lim_{a\to\infty} \left[\frac{1}{9} \times \frac{1}{\frac{2}{3}} \times \arctan\frac{2}{3}x\right]_0^a$

$= \lim_{a\to\infty} \left[\frac{1}{6}\arctan\frac{2}{3}x\right]_0^a$

$= \lim_{a\to\infty} \left(\frac{1}{6}\arctan\frac{2}{3}a - \frac{1}{6}\arctan 0\right)$

$= \lim_{a\to\infty} \left(\frac{1}{6}\arctan\frac{2}{3}a\right)$

As $a \to \infty$, $\arctan\frac{2}{3}a \to \frac{\pi}{2}$

$\therefore \int_0^\infty \frac{1}{4x^2+9}\,\mathrm{d}x = \frac{1}{6} \times \frac{\pi}{2} = \frac{\pi}{12}$

30)

$\frac{12-6x}{(2x+3)(x^2+3)} \equiv \frac{A}{2x+3} + \frac{Bx+C}{x^2+3}$

$\Rightarrow 12 - 6x \equiv A(x^2 + 3) + (Bx + C)(2x + 3)$

Let $x = -\frac{3}{2}$: $21 = \frac{21}{4}A \Rightarrow A = 4$

Let $x = 0$: $12 = 12 + 3C \Rightarrow C = 0$

Let $x = 1$: $6 = 16 + 5B \Rightarrow B = -2$

$A = 4, B = -2, C = 0$

$\therefore \frac{12-6x}{(2x+3)(x^2+3)} \equiv \frac{4}{2x+3} - \frac{2x}{x^2+3}$

$\int_0^\infty \frac{12-6x}{(2x+3)(x^2+3)} \mathrm{d}x$

$= \lim_{a\to\infty} \int_0^a \left(\frac{4}{2x+3} - \frac{2x}{x^2+3}\right) \mathrm{d}x$

$= \lim_{a\to\infty} [2\ln|2x + 3| - \ln|x^2 + 3|]_0^a$

$= \lim_{a\to\infty} \left((2\ln|2a + 3| - \ln|a^2 + 3|) - (2\ln|2 \times 0 + 3| - \ln|0^2 + 3|)\right)$

$= \lim_{a\to\infty} \left((\ln|(2a + 3)^2| - \ln|a^2 + 3|) - (2\ln 3 - \ln 3)\right)$

$= \lim_{a\to\infty} \left(\ln\left|\frac{(2a+3)^2}{a^2+3}\right| - \ln 3\right)$

$\frac{(2a+3)^2}{a^2+3} \equiv \frac{4a^2+12a+9}{a^2+3} \equiv \frac{4+\frac{12}{a}+\frac{9}{a^2}}{1+\frac{3}{a^2}}$

As $a \to \infty$, $\frac{(2a+3)^2}{a^2+3} \to 4$, so $\ln\left|\frac{(2a+3)^2}{a^2+3}\right| \to \ln 4$

$\therefore \int_0^\infty \frac{12-6x}{(2x+3)(x^2+3)} \mathrm{d}x = \ln 4 - \ln 3 = \ln\frac{4}{3}$

So $k = \frac{4}{3}$

31)

$\frac{3x^2+16}{x^2(x^2+4)} \equiv \frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+4}$

$\Rightarrow 3x^2 + 16 \equiv Ax(x^2+4) + B(x^2+4) + (Cx+D)x^2$

Let $x = 0$: $16 = 4B \Rightarrow B = 4$

Let $x = 1$: $19 = 5A + 20 + C + D \Rightarrow 5A + C + D = -1$

Let $x = -1$: $19 = -5A + 20 - C + D \Rightarrow -5A - C + D = -1$

Let $x = 2$: $28 = 16A + 32 + 8C + 4D \Rightarrow 16A + 8C + 4D = -4$

$A = 0$, $B = 4$, $C = 0$, $D = -1$

$\therefore \frac{3x^2+16}{x^2(x^2+4)} \equiv \frac{4}{x^2} - \frac{1}{x^2+4}$

$\int_2^\infty \frac{3x^2+16}{x^2(x^2+4)}\,\mathrm{d}x$

$= \lim_{a\to\infty} \int_2^a \left(\frac{4}{x^2} - \frac{1}{4+x^2}\right)\mathrm{d}x$

$= \lim_{a\to\infty} \int_2^a \left(4x^{-2} - \frac{1}{4\left(1+\frac{1}{4}x^2\right)}\right)\mathrm{d}x$

$= \lim_{a\to\infty} \int_2^a \left(4x^{-2} - \frac{1}{4\left(1+\left(\frac{1}{2}x\right)^2\right)}\right)\mathrm{d}x$

$= \lim_{a\to\infty} \left[-4x^{-1} - \frac{1}{4} \times \frac{1}{\frac{1}{2}} \times \arctan\frac{1}{2}x\right]_2^a$

$= \lim_{a\to\infty} \left[-\frac{4}{x} - \frac{1}{2}\arctan\frac{1}{2}x\right]_2^a$

$= \lim_{a\to\infty} \left(\left(-\frac{4}{a} - \frac{1}{2}\arctan\frac{1}{2}a\right) - \left(-2 - \frac{1}{2}\arctan 1\right)\right)$

$= \lim_{a\to\infty} \left(-\frac{4}{a} - \frac{1}{2}\arctan\frac{1}{2}a + 2 + \frac{\pi}{8}\right)$

As $a \to \infty$, $\frac{4}{a} \to 0$ and $\arctan\frac{1}{2}a \to \frac{\pi}{2}$

$\therefore \int_2^\infty \frac{3x^2+16}{x^2(x^2+4)}\,\mathrm{d}x = -\frac{1}{2} \times \frac{\pi}{2} + 2 + \frac{\pi}{8}$

$= 2 - \frac{\pi}{8}$

So $k = \frac{1}{8}$

32)

$\int_1^2 \frac{1}{\sqrt{x^2-1}}\,\mathrm{d}x$

$= \lim_{a\to 1} \int_a^2 \frac{1}{\sqrt{x^2-1}}\,\mathrm{d}x$

$= \lim_{a\to 1} [\operatorname{arcosh} x]_a^2$

$= \lim_{a\to 1} (\operatorname{arcosh} 2 - \operatorname{arcosh} a)$

$= \lim_{a\to 1} \left(\ln\left(2+\sqrt{2^2-1}\right) - \ln\left(a+\sqrt{a^2-1}\right)\right)$

$= \lim_{a\to 1} \left(\ln\left(2+\sqrt{3}\right) - \ln\left(a+\sqrt{a^2-1}\right)\right)$

As $a \to 1$, $\ln\left(a+\sqrt{a^2-1}\right) \to 0$

$\therefore \int_1^2 \frac{1}{\sqrt{x^2-1}}\,\mathrm{d}x = \ln\left(2+\sqrt{3}\right)$

33)

$\int_{-\infty}^{\infty} \frac{1}{x^2+8x+17}\,\mathrm{d}x$

$= \int_{-\infty}^{\infty} \frac{1}{(x+4)^2-16+17}\,\mathrm{d}x$

$= \int_{-\infty}^{\infty} \frac{1}{(x+4)^2+1}\,\mathrm{d}x = \int_{-\infty}^{-4} \frac{1}{(x+4)^2+1}\,\mathrm{d}x + \int_{-4}^{\infty} \frac{1}{(x+4)^2+1}\,\mathrm{d}x$

$= \lim_{a\to-\infty} \int_a^{-4} \frac{1}{(x+4)^2+1}\,\mathrm{d}x + \lim_{a\to\infty} \int_{-4}^{\infty} \frac{1}{(x+4)^2+1}\,\mathrm{d}x$

$= \lim_{a\to-\infty} [\arctan(x+4)]_a^{-4} + \lim_{a\to\infty} [\arctan(x+4)]_{-4}^{a}$

$= \lim_{a\to-\infty} (\arctan 0 - \arctan(a+4)) + \lim_{a\to\infty} (\arctan(a+4) - \arctan 0)$

$= \lim_{a\to-\infty} (-\arctan(a+4)) + \lim_{a\to\infty} (\arctan(a+4))$

As $a \to -\infty$, $\arctan(a+4) \to -\frac{\pi}{2}$

As $a \to \infty$, $\arctan(a+4) \to \frac{\pi}{2}$

$\therefore \int_{-\infty}^{\infty} \frac{1}{x^2+8x+17}\,\mathrm{d}x = -\left(-\frac{\pi}{2}\right) + \frac{\pi}{2}$

$= \pi$

34)

$\int_0^1 \frac{\ln x}{\sqrt{x}}\,\mathrm{d}x$

$= \lim_{a\to 0} \int_a^1 \frac{\ln x}{\sqrt{x}}\,\mathrm{d}x$

Use Integration by Parts:

$u = \ln x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = x^{-\frac{1}{2}}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x} \qquad v = 2x^{\frac{1}{2}}$

$\therefore \int \frac{\ln x}{\sqrt{x}}\,\mathrm{d}x = 2\sqrt{x}\ln x - \int 2x^{-\frac{1}{2}}\,\mathrm{d}x$

$= 2\sqrt{x}\ln x - 4\sqrt{x} + c$

$\therefore \lim_{a\to 0} \int_a^1 \frac{\ln x}{\sqrt{x}}\,\mathrm{d}x$

$= \lim_{a\to 0} \left[2\sqrt{x}\ln x - 4\sqrt{x}\right]_a^1$

$= \lim_{a\to 0} \left(\left(2\sqrt{1}\ln 1 - 4\sqrt{1}\right) - \left(2\sqrt{a}\ln a - 4\sqrt{a}\right)\right)$

$= \lim_{a\to 0} \left(-4 - 2\sqrt{a}\ln a + 4\sqrt{a}\right)$

As $a \to 0$, $\sqrt{a}\ln a \to 0$

Note that this can be justified using L'Hôpital's Rule,

e.g. $\lim_{a\to 0} \sqrt{a}\ln a = \lim_{a\to 0} \frac{\ln a}{a^{-\frac{1}{2}}} = \lim_{a\to 0} \frac{\frac{1}{a}}{-\frac{1}{2}a^{-\frac{3}{2}}} = \lim_{a\to 0} \left(\frac{1}{a} \times -2a^{\frac{3}{2}}\right) = \lim_{a\to 0} \left(-2\sqrt{a}\right) = 0$

$\therefore \int_0^1 \frac{\ln x}{\sqrt{x}}\,\mathrm{d}x = -4$

35)

$\int_0^{12} \ln x \, \mathrm{d}x$

$= \lim_{a \to 0} \int_a^{12} 1 \times \ln x \, \mathrm{d}x$

Use Integration by Parts:

$u = \ln x$ $\qquad$ $\frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x}$ $\qquad$ $v = x$

$\therefore \int \ln x \, \mathrm{d}x = x \ln x - \int 1 \, \mathrm{d}x$

$= x \ln x - x + c$

$\therefore \lim_{a \to 0} \int_a^{12} \ln x \, \mathrm{d}x$

$= \lim_{a \to 0} [x \ln x - x]_a^{12}$

$= \lim_{a \to 0} \big((12 \ln 12 - 12) - (a \ln a - a)\big)$

As $a \to 0$, $a \ln a \to 0$

Note that this can be justified using L'Hôpital's Rule,

e.g. $\lim_{a \to 0} a \ln a = \lim_{a \to 0} \frac{\ln a}{\frac{1}{a}} = \lim_{a \to 0} \frac{\frac{1}{a}}{-\frac{1}{a^2}} = \lim_{a \to 0} (-a) = 0$

$\therefore \int_0^{12} \ln x \, \mathrm{d}x = 12 \ln 12 - 12$

36)

$\int_0^\infty x^2 \mathrm{e}^{-3x}\,\mathrm{d}x$

$= \lim_{a\to\infty} \int_0^a x^2 \mathrm{e}^{-3x}\,\mathrm{d}x$

Use Integration by Parts:

$$u = x^2 \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{-3x}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = 2x \qquad v = -\frac{1}{3}\mathrm{e}^{-3x}$$

$\therefore \int x^2 \mathrm{e}^{-3x}\,\mathrm{d}x = -\frac{1}{3}x^2 \mathrm{e}^{-3x} + \int \frac{2}{3} x \mathrm{e}^{-3x}\,\mathrm{d}x$

Use Integration by Parts:

$$u = \frac{2}{3}x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{-3x}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2}{3} \qquad v = -\frac{1}{3}\mathrm{e}^{-3x}$$

$\therefore \int x^2 \mathrm{e}^{-3x}\,\mathrm{d}x = -\frac{1}{3}x^2 \mathrm{e}^{-3x} - \frac{2}{9} x \mathrm{e}^{-3x} + \int \frac{2}{9} \mathrm{e}^{-3x}\,\mathrm{d}x$

$= -\frac{1}{3}x^2 \mathrm{e}^{-3x} - \frac{2}{9} x \mathrm{e}^{-3x} - \frac{2}{27}\mathrm{e}^{-3x} + c$

$\therefore \lim_{a\to\infty} \int_0^a x^2 \mathrm{e}^{-3x}\,\mathrm{d}x$

$= \lim_{a\to\infty} \left[-\frac{1}{3}x^2 \mathrm{e}^{-3x} - \frac{2}{9} x \mathrm{e}^{-3x} - \frac{2}{27}\mathrm{e}^{-3x}\right]_0^a$

$= \lim_{a\to\infty} \left(\left(-\frac{1}{3}a^2 \mathrm{e}^{-3a} - \frac{2}{9} a \mathrm{e}^{-3a} - \frac{2}{27}\mathrm{e}^{-3a}\right) - \left(-\frac{1}{3}\times 0 \times \mathrm{e}^0 - \frac{2}{9}\times 0 \times \mathrm{e}^0 - \frac{2}{27}\mathrm{e}^0\right)\right)$

$= \lim_{a\to\infty} \left(-\frac{1}{3}a^2 \mathrm{e}^{-3a} - \frac{2}{9} a \mathrm{e}^{-3a} - \frac{2}{27}\mathrm{e}^{-3a} + \frac{2}{27}\right)$

As $a \to \infty$, $a^2 \mathrm{e}^{-3a} \to 0$ and $a\mathrm{e}^{-3a} \to 0$ and $\mathrm{e}^{-3a} \to 0$

Note that this can be justified using L'Hôpital's Rule,

e.g. $\lim_{a\to\infty} a^2 \mathrm{e}^{-3a} = \lim_{a\to\infty} \frac{a^2}{\mathrm{e}^{3a}} = \lim_{a\to\infty} \frac{2a}{3\mathrm{e}^{3a}} = \lim_{a\to\infty} \frac{2}{9\mathrm{e}^{3a}} = 0$

$\therefore \int_0^\infty x^2 \mathrm{e}^{-3x}\,\mathrm{d}x = \frac{2}{27}$

37) a)

$\lim_{x\to\infty} \left(\frac{\ln x}{x}\right) = \lim_{x\to\infty} \left(\frac{\frac{1}{x}}{1}\right) = \lim_{x\to\infty} \left(\frac{1}{x}\right) = 0$

37) b)

$\int_1^{\infty} \frac{\ln x}{x^2}\,\mathrm{d}x$

$= \lim_{a\to\infty} \int_1^{a} \frac{\ln x}{x^2}\,\mathrm{d}x$

Use Integration by Parts:

$u = \ln x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = x^{-2}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x} \qquad v = -x^{-1}$

$\therefore \int \frac{\ln x}{x^2}\,\mathrm{d}x = -x^{-1}\ln x + \int x^{-2}\,\mathrm{d}x$

$= -\frac{\ln x}{x} - x^{-1} + c$

$\therefore \int_1^{\infty} \frac{\ln x}{x^2}\,\mathrm{d}x = \lim_{a\to\infty}\left[-\frac{\ln x}{x} - \frac{1}{x}\right]_1^{a}$

$= \lim_{a\to\infty}\left(\left(-\frac{\ln a}{a} - \frac{1}{a}\right) - \left(-\frac{\ln 1}{1} - \frac{1}{1}\right)\right)$

$= \lim_{a\to\infty}\left(-\frac{\ln a}{a} - \frac{1}{a} + 1\right)$

$= \lim_{a\to\infty}\left(-\frac{\ln a}{a}\right) + \lim_{a\to\infty}\left(-\frac{1}{a} + 1\right)$

From part a):

$= 0 + \lim_{a\to\infty}\left(-\frac{1}{a} + 1\right)$

As $a \to \infty$, $\frac{1}{a} \to 0$

$= 0 + 1 = 1$

$\therefore \int_1^{\infty} \frac{\ln x}{x^2}\,\mathrm{d}x = 1$

38) a)

$\lim_{x\to\infty}\left(\frac{x}{\mathrm{e}^x}\right) = \lim_{x\to\infty}\left(\frac{1}{\mathrm{e}^x}\right) = 0$

38) b)

$\int_{-\infty}^{0} x\mathrm{e}^{x}\,\mathrm{d}x$

$= \lim_{a\to-\infty} \int_{a}^{0} x\mathrm{e}^{x}\,\mathrm{d}x$

Use Integration by Parts:

$u = x$ $\qquad$ $\frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{x}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = 1$ $\qquad$ $v = \mathrm{e}^{x}$

$\therefore \int x\mathrm{e}^{x}\,\mathrm{d}x = x\mathrm{e}^{x} - \int \mathrm{e}^{x}\,\mathrm{d}x$

$= x\mathrm{e}^{x} - \mathrm{e}^{x} + c$

$\therefore \int_{-\infty}^{0} x\mathrm{e}^{x}\,\mathrm{d}x = \lim_{a\to-\infty} [x\mathrm{e}^{x} - \mathrm{e}^{x}]_{a}^{0}$

$= \lim_{a\to-\infty} \left((0 \times \mathrm{e}^{0} - \mathrm{e}^{0}) - (a\mathrm{e}^{a} - \mathrm{e}^{a})\right)$

$= \lim_{a\to-\infty} (-1 - a\mathrm{e}^{a} + \mathrm{e}^{a})$

As $a \to -\infty$, $\mathrm{e}^{a} \to 0$

$a\mathrm{e}^{a} = \frac{a}{\mathrm{e}^{-a}}$

Let $-a = u \Rightarrow \frac{a}{\mathrm{e}^{-a}} = -\frac{u}{\mathrm{e}^{u}}$

As $a \to -\infty$, $u \to \infty$ so $-\frac{u}{\mathrm{e}^{u}} \to 0$ so $a\mathrm{e}^{a} \to 0$

$\therefore \int_{-\infty}^{0} x\mathrm{e}^{x}\,\mathrm{d}x = -1$

39)

$\int_3^\infty \frac{\ln x - 1}{(\ln x)^2}\,\mathrm{d}x$

$= \lim_{a\to\infty} \int_3^a \frac{\ln x - 1}{(\ln x)^2}\,\mathrm{d}x$

$= \lim_{a\to\infty} \int_3^a \left(\frac{1}{\ln x} - \frac{1}{(\ln x)^2}\right)\mathrm{d}x$

Consider $I = \int \frac{1}{\ln x}\,\mathrm{d}x$

Use Integration by Parts:

$$u = \frac{1}{\ln x} \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{\ln x \times 0 - 1 \times \frac{1}{x}}{(\ln x)^2} \qquad v = x$$

$$= -\frac{1}{x(\ln x)^2}$$

$\therefore I = \int \frac{1}{\ln x}\,\mathrm{d}x = \frac{x}{\ln x} + \int \frac{1}{(\ln x)^2}\,\mathrm{d}x$

$\Rightarrow \int \frac{1}{\ln x}\,\mathrm{d}x - \int \frac{1}{(\ln x)^2}\,\mathrm{d}x = \frac{x}{\ln x} + c$

$\Rightarrow \int \left(\frac{1}{\ln x} - \frac{1}{(\ln x)^2}\right)\mathrm{d}x = \frac{x}{\ln x} + c$

$\therefore \lim_{a\to\infty} \int_3^a \left(\frac{1}{\ln x} - \frac{1}{(\ln x)^2}\right)\mathrm{d}x = \lim_{a\to\infty} \left[\frac{x}{\ln x}\right]_3^a$

$= \lim_{a\to\infty} \left(\frac{a}{\ln a} - \frac{3}{\ln 3}\right)$

Use L'Hôpital's Rule:

$\lim_{a\to\infty} \left(\frac{a}{\ln a}\right) = \lim_{a\to\infty} \left(\frac{1}{\frac{1}{a}}\right) = \lim_{a\to\infty} a$ which diverges

So $\int_3^\infty \frac{\ln x - 1}{(\ln x)^2}\,\mathrm{d}x$ diverges.